Dr. Jinxin Li

Physicist & Scientific Software Developer

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PROFILE

Physicist and scientific software developer working with C++, Python and AI agent development. Experience in Linux/HPC software, algorithm development, framework maintenance, debugging, integration and automated validation. Technical owner and primary maintainer of the existing LuminosityFit framework during doctoral work, and active PandaRoot contributor. Architect and developer of PANDA Agent using an AI-assisted workflow, with responsibility for architecture, retrieval design, evaluation, failure analysis and technical acceptance.

SELECTED ENGINEERING PROJECTS

PANDA Agent — Evidence-grounded Scientific Code & Literature QA

Apr. 2026 – Present · Local research prototype · Personal project · github.com/JinxinLee/PANDA_Agent

- Architected and developed a version-aware research assistant connecting code, papers and documentation through hybrid retrieval, evidence sufficiency review, claim verification and bounded revision. Used an AI-assisted workflow, retaining ownership of requirements, architecture, failure analysis, review and technical acceptance.
- Designed human-reviewed evaluation datasets with separate development and validation sets and an externally managed sealed holdout.
- **Evaluation results — Attempt 5:** 97.96% Gold recall@10; 94.92% answer-point coverage; 100% citation integrity; 4.62% identifier hallucination rate.

LuminosityFit — C++ Framework Development

Mar. 2022 – Jul. 2024 · Doctoral research

- Served as technical owner and primary maintainer of an existing scientific C++ framework.
- Developed extended-target and beam-property fitting workflows; debugged and validated changes and integrated functionality with PandaRoot.
- Recovered input luminosity within 0.5% relative deviation in the studied simulation scenarios.

Reconstruction & Iterative Correction

Jul. 2024 – Apr. 2026 · Doctoral research

- Developed reconstruction workflows and simulation-based efficiency corrections; integrated existing tracking and GenFit/Kalman components.
- Reduced a deliberately introduced 40% profile-model error to a 1.3% global discrepancy within three iterations in the documented simulation study.

Detector Simulation & Performance Evaluation

Sep. 2019 – Mar. 2022 · Doctoral research

- Designed and implemented simulated HV-MAPS detector geometry in PandaRoot.
- Benchmarked tracking, vertex/momentum resolution and event reconstruction against the baseline detector design.

EXPERIENCE

Doctoral Researcher / Scientific Software Developer

Ruhr University Bochum · Institute of Experimental Physics I · PANDA Collaboration
Sep. 2019 – Jul. 2026

- Developed and maintained C++ and Python software for detector reconstruction, scientific data processing and statistical analysis.
- Served as technical owner / primary maintainer of the existing LuminosityFit framework and Software Owner for the PANDA Luminosity Detector Group.
- Contributed functionality, reconstruction changes, bug fixes and cross-component integration to PandaRoot.
- Developed automated validation workflows and technical documentation; provided training and undergraduate thesis mentoring.

SKILLS

Engineering: C++, Python, Bash/Shell, ROOT, PandaRoot, FairRoot, Git, Linux; software maintenance, debugging and integration.

Computing: SLURM, Docker, Singularity, NumPy, Pandas, Scikit-learn.

Methods: Monte Carlo simulation, statistical data analysis, detector tracking and vertex reconstruction, automated testing and validation.

AI architecture & evaluation: AI agent / RAG workflow architecture, hybrid retrieval, citation grounding, benchmark design and failure analysis.

Languages: Chinese (Native), English (Proficient), German (CEFR A2).

EDUCATION

Dr. rer. nat. in Physics — Ruhr-Universität Bochum

Germany · Sep. 2019 – Jul. 2026

Dissertation: “Luminosity Determination with Restgas Background from the Target for the PANDA Experiment”.

B.Sc. in Physics — Nankai University

Tianjin, China · Sep. 2015 – Jun. 2019

EARLY RESEARCH EXPERIENCE

IHEP, Chinese Academy of Sciences / BESIII Jun. 2018 – Jul. 2019
Research Assistant · Beijing, China. Charm/D-meson analysis, Monte Carlo simulation and ROOT-based statistics.

Nankai University / BESIII Sep. 2016 – Jul. 2019
Research Assistant / Undergraduate Research Assistant · Tianjin, China. Charmonium analysis and simulation.

CERN / MIT / AMS Jun. 2017 – Sep. 2017
Research Assistant / Summer Research Assistant · Geneva, Switzerland. AMS-02 simulation and tracker-resolution validation.